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يُونِيسَيتِي اِسْلَامُ اِنْتَارَا بَغْسِيَا مِلْدِسِيَا
Garden of Knowledge and Virtue

KULLIYAH OF ENGINEERING

PROJECT GROUP

Assessment 1:

LLMS-CANT-PLEASE-THEM-ALL

MCTA 4363 DEEP LEARNING

Section 1

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1. Introduction & Problem Statement

The rapid advancement of Large Language Models (LLMs) has enabled machines to generate highly coherent and human-like text. However, evaluating such generated content remains challenging due to the subjective nature of human preferences and the non-deterministic architecture of LLMs.

This project aligns with the Kaggle competition "LLMs: You Can't Please Them All" , which focuses on modeling disagreements among evaluators when assessing AI-generated responses. The core problem addressed in this assessment is how to consistently generate text that receives diverse, conflicting evaluations from a panel of judges, and how to effectively quantify that disagreement.

2. Core Objectives

The primary focus of Assessment 1 is to establish a foundational, offline baseline for our Adversarial NLP framework. The objectives are:

- **Develop an Adversarial Framework:** Create a system capable of exploiting vulnerabilities in an LLM-as-a-judge setup.
- **Maximize Disagreement:** Generate essays that maximize the scoring variance (disagreement) between a simulated committee of three distinct LLM judges.
- **Understand LLM Limitations:** Form a better understanding of the capabilities and limitations of using LLMs for subjective evaluation tasks at scale.

3. Methodology & Approach

To achieve these objectives, this baseline project uses a **prompt-based essay generation approach** designed to intentionally exploit differences in LLM evaluation.

3.1 Writing Strategies

Instead of generating standard, balanced academic essays, the system employs three specific writing strategies designed to trigger scoring anomalies:

- **Philosophical:** Utilizes abstract reasoning, ethical ambiguity, and contradictory ideas to confuse rigid logical evaluators.
- **Metaphorical:** Employs rich imagery and layered meanings, which tend to be rewarded by creative evaluators but penalized by strict, rule-based evaluators.
- **Dialectical:** Presents competing viewpoints (Thesis vs. Antithesis) to create logical tension without providing a definitive resolution.

3.2 Offline System Architecture

For Assessment 1, **no live APIs were used**. Both the Generator and the Judge Committee function as simulated, offline models to establish a controlled baseline and avoid API costs or external network latency during initial testing.

- **The Generator (Offline Model):** * Conceptually based on a transformer-based language model (e.g., GPT or Llama-3.1-8B).
 - In this offline phase, the generator acts as an Adversarial Agent by utilizing a randomized template-matching system. It ingests an input topic and constructs a "payload" essay (~100 words in English) by injecting the topic into predefined structural templates representing the three strategies.
- **The Judge Committee (Offline Simulation):** * A simulated committee of three hidden LLM evaluators (conceptually representing models like Gemma-2, Llama-3, and GPT-Neo).
 - Because LLMs are non-deterministic and can produce "noise" (e.g., scoring a 7 one time and a 6 the next), this offline simulation uses statistical randomization (**numpy** and **random** libraries) to assign scores across five criteria. This allows us to safely benchmark the mathematical logic of our pipeline before moving to live models.

4. Pipeline Execution

The system operates through a sequential, multi-agent adversarial pipeline:

1. **Topic Input:** The system loads a diverse list of essay topics (e.g., healthcare adaptability, management consulting).

2. **Adversarial Generation:** The offline Generator selects a strategy (Philosophical, Metaphorical, or Dialectical) and generates a unique candidate essay.
3. **Multi-Judge Evaluation:** The generated essay is passed to the simulated Judge Committee. Each judge assigns a score (2.0 to 10.0) across five criteria: *Introduction, Logic, Emotion, Philosophy, and Conclusion*.
4. **Metric Calculation:** The system computes the disagreement.
5. **Score Collection & Export:** Results are tabulated, ranked, and exported to CSV format ([submission.csv](#) and [results.csv](#)) for further analysis.

5. Metrics & Evaluation Goal

The system evaluates success not by how "correct" or "perfect" an essay is, but by the level of disagreement it causes.

For this baseline, the scoring layer tracks:

- **Average Quality (avg_q):** The mean score given by the judges across all criteria.
- **Average Disagreement (avg_h/stddev):** The standard deviation of the judges' scores, indicating the horizontal variance or disagreement level.

The **Final System Score** is calculated using the following mathematical logic:

Final Score = Average Disagreement × Minimum Disagreement × (9 - Average Quality)

This equation ensures that the highest-ranking essays are those that maintain an acceptable quality baseline while maximizing confusion and polarized scoring among the committee members.

6. Preliminary Results

A controlled offline trial was conducted using three distinct topics. The system successfully generated essays and simulated multi-judge disagreement.

Output Summary

Essay Topic	Avg Quality Score	Avg Disagreement	Final System Score
1. Compare the importance of self-reliance and adaptability in healthcare.	6.22 / 10.0	2.50	14.15

2. Evaluate management consulting in addressing conflicts within marketing.	5.86 / 10.0	1.83	3.51
3. Discuss the role of self-reliance in achieving success in software engineering.	6.13 / 10.0	2.11	9.07

Analysis of the Best Performing Essay (Essay 1)

- **Strategy Deployed:** Metaphorical
- **Excerpt:** *"Think of compare the importance of self-reliance and adaptability in healthcare as a river and a dam. The river — wild and indifferent — represents raw potential... Some may call this progress, while others may call it loss."*
- **Result:** This essay achieved the highest **Final Score (14.15)** because the metaphorical approach caused massive divergence in the simulated judges' rubrics. While the average quality remained acceptable (6.22), the standard deviation between the scores was heavily polarized (2.50), successfully fulfilling the project objective.

7. Conclusion

Assessment 1 successfully established an end-to-end evaluation workflow. By utilizing offline simulated models, we validated our mathematical metrics and adversarial prompt logic without incurring API costs. The pipeline successfully generated diverse essay candidates and demonstrated that strategic ambiguity (like metaphors and paradoxes) reliably increases scoring variance among evaluators.